# Public Disclosure, Trading Costs, and International Policy Diffusion

*Evidence from global reporting regimes*


## Abstract

Reporting rules differ in the information they require, the firms they cover and the date on which investors can obtain the report. We examine these distinctions across global disclosure policies using 1,055,142 security-years from 90 markets during 2000-2025. The clearest descriptive pattern is spread compression around the initial public banking country-by-country reporting phase in Europe. Country-market spreads are 21.9% lower relative to the comparison markets with equal-country weights and 10.3% lower with predetermined GDP weights. The pattern does not extend uniformly to price impact or the reviewed mandatory E/S/G phases. Banking disclosure is one coordinated regulatory wave, and the comparisons do not identify effects on eligible bank issuers. International timing models distinguish competition, learning, emulation and financial dependence; no focal coefficient survives correction across 433 stable coefficients. The evidence identifies a specific spread association and shows why scope, information timing and collective implementation must be separated before attributing capital-market benefits or diffusion mechanisms to disclosure mandates.



Keywords: public disclosure; stock liquidity; banking transparency; mandatory ESG; international diffusion


## 1 Introduction

Disclosure regulation seeks to change the information available to market participants. Whether it lowers trading costs depends on what becomes public, which firms supply the information and when the obligation takes effect. A rule requiring environmental policies, a confidential tax return and a public report of banking activities can therefore have different capital-market consequences even when each is described as a disclosure reform.

We examine two connected questions. First, how does market liquidity change around the implementation of reporting requirements with different information content and audiences? Second, is domestic implementation associated with earlier policies or economic outcomes among foreign peers? The first question concerns the information environment; the second concerns the process through which regulation spreads.

Our strongest market pattern concerns the initial European public banking country-by-country reporting phase in 2014. The equal-country log-spread comparison is -0.247, and the constant-GDP-weighted comparison is -0.109. Both indicate lower spreads. The direction remains negative across the six predetermined weighting schemes, although the magnitude varies substantially. The result concerns the equity markets of EU home jurisdictions; it is not an estimate for a verified sample of legally eligible banks.

This distinction disciplines the interpretation. GDP-weighted price impact rises around the banking phases, the full-reporting spread comparison changes sign after a pretrend adjustment, and reviewed ESG phases do not show a consistent liquidity improvement. We therefore develop a focused account of the observed spread compression rather than infer a general benefit from every disclosure obligation. Koning et al. (2018) motivate our separate tests of diffusion mechanisms; the banking wave itself cannot supply independent national adoption decisions.

Krueger et al. (2023) examine mandatory ESG reporting and its liquidity consequences. We extend the economic comparison across reporting obligations with different content and audiences. Separating comprehensive E/S/G responses, climate-only requirements, confidential tax reporting and public banking disclosures makes the relevant information channel explicit. Our evidence concerns policy environments rather than the eligible-firm effects studied in their design.

## 2 Institutional Setting and Predictions

Public information can reduce adverse selection when it supplies credible, incremental facts that investors otherwise obtain unevenly. Lower adverse selection predicts narrower spreads and lower price impact. Disclosure that repeats existing information, permits superficial responses or imposes costs without improving information need not generate the same response. We predict stronger direct-information effects when the relevant content is publicly accessible and incremental, while treating the direction of each observed outcome as an empirical question.

Mandatory reporting is different from mandatory conduct. A comply-or-explain rule requires a response even if the firm does not follow a recommended practice. A requirement to describe CSR spending does not by itself establish comprehensive environmental, social and governance reporting. Following Krueger et al., a full E/S/G assignment requires all three topics for the relevant issuer population; the last topic determines the joint date. Their voluntary firm-disclosure proxy is I/B/E/S earnings guidance, not voluntary ESG reporting.

Banking public CbCR has two information phases. Article 89 of Directive 2013/36/EU initially requires public names, activities, locations, turnover and employees by July 2014. From 2015, annual reporting adds pretax profits, taxes and subsidies. The 2014 profit-and-tax submission by globally systemic institutions is confidential and is not the full public phase. Confidential BEPS Action 13 and general multinational public income-tax reporting are separate regimes. The banking rule does not use the general EUR750 million multinational revenue threshold.

Eligibility follows the regulated institution and its consolidation perimeter. UK guidance excludes holding companies as institutions and distinguishes third-country branches from authorised subsidiaries; French law expressly includes certain financial holding entities. A listed parent can publish a group report without itself being the regulated filer. These national differences preclude treating every EU-listed equity as an eligible bank.

International diffusion generates separate predictions. Competition concerns earlier implementation among export competitors. Learning concerns prior favourable outcomes among earlier implementers. Emulation concerns policy choices among institutional peers. Coercion requires external dependence and influence over the particular policy. These are distinct explanations; an association with peer exposure or IMF credit does not by itself identify the underlying mechanism.

## 3 Data and Research Design

### 3.1 Reporting scope and timing

The policy register contains 441 event records. The ESG legal review covers 41 countries, including all 39 countries in the author-released ESG workbook. It identifies 37 qualifying comprehensive reporting phases and withholds full-ESG assignments for Argentina, Canada, Pakistan and China under the stated definition. These exclusions reflect subnational scope, conditional financial disclosure, CSR-activity reporting or an incompletely verified full-topic template; they do not establish that those countries have no ESG-related obligations.

Table 1 separates the principal obligations. France explicitly applies the NRE environmental and social paragraph from fiscal years beginning in 2002, while its remuneration and director-mandate provisions start in 2001. Hong Kong requires E/S general disclosures from fiscal years starting in 2016, with environmental KPIs following in 2017. Singapore first covers years ending on or after 31 December 2017. India initially covers the largest 100 listed issuers in 2012; that population differs from the later largest-500 expansion used in the reference paper.

National implementation matters within Europe. Ireland specifies financial years beginning on or after 1 August 2017, Finland refers to years ending on 31 December 2017, and the Greek implementation circular covers years ending on or after 7 July 2016. The reviewed phases are not an exhaustive census of the earliest separate national E, S and G duties. Some national commencement clauses remain supported by official summaries or country mapping rather than a fully archived original. Evidence grades and gaps accompany each assignment.

Historical EBA supervisory rosters identify 123 groups in 2014 and 105 in 2015, with 143 distinct LEIs across the two lists. They are supervisory samples, not a complete Article 89 filer census. Five archived HSBC, Deutsche Bank and BNP Paribas Fortis report examples corroborate partial and full public content. HSBC dates publication of its 2014 CbCR report to 30 June 2015. Other report-period ends and PDF file-name dates are not substituted for observed publication dates. A Deutsche Bank 2014 annual report corroborates the issuer ISIN used in a provisional vendor link; a complete historical bank eligibility panel remains unavailable.

**Table 1 Principal reporting obligations and covered populations**

| Regime | Information and audience | Covered population / timing |
|---|---|---|
| Reviewed mandatory E/S/G | Public topic-specific responses; comply-or-explain counts as required reporting. | 37 reviewed country phases. Listing, size, sector and materiality rules apply. These are not certified first-ever national ESG dates. |
| Bank public CbCR: initial | Names, activities, locations, turnover and employees. | Article 89 regulated institutions; public by July 2014. The 28 EU home-jurisdiction entries are one collective wave. |
| Bank public CbCR: full | Adds pretax profit, tax and public subsidies. | Annual full reporting from 2015; fiscal periods and national filer rules differ. Same collective wave; no EUR 750m threshold. |
| Confidential tax CbCR | Multinational tax information to tax administrations. | BEPS Action 13 thresholds, parent/surrogate/local filing roles. Not public banking CbCR. |
| Financial reporting (IFRS) | Public financial statements under applicable standards. | Listed/consolidated and national population rules; transitional phases differ. |

Dates identify the reviewed reporting phase and population, not an unverified global first-topic census. General multinational public tax reporting, extractive payments and climate-only rules remain separate classes in the companion register. No future schedule is included. All 18 class counts and country rows are online; 444 country-class phase entries through 2025 include coordinated bank home-geography entries, not 444 independent adoptions.

### 3.2 Market outcomes and comparisons

The Datastream panel contains 99,100 quote identifiers across 90 jurisdictions during 2000-2025. Major equities and primary country quotes are screened for duplicate or conflicting daily observations, delisting, stale prices and extreme-return reversals. Annual outcomes include log quoted spread, log Amihud price impact and the zero-return fraction, each with at least 120 valid daily observations. Both logged proxies use within-year 1/99 winsorisation. Secondary outcomes are an illiquidity principal component, trading value relative to market capitalisation and return volatility.

For a phase assigned to calendar year g, we compare the same securities in g-1 and g+1, require at least 30 balanced securities per country, and subtract changes in at least three comparison countries. ESG comparisons use later reviewed phases; an unresolved history is not coded as never treated. Bank comparisons use the EU home-jurisdiction geography against non-EU home jurisdictions, excluding non-EU EEA countries. Outside the EU does not imply absence of other reporting or exposure through European subsidiaries.

Country weights are equal, lagged constant-dollar GDP, current-dollar GDP, purchasing-power GDP, sampled market capitalisation or balanced quote counts. GDP-matched equal-weight cells separate weighting from data availability. Prespecified diagnostics vary public-availability timing, pretrend extrapolation, post-period averaging, quote weighting, common-outcome samples, trading-day support, size screens, winsorisation, independent-case support and concurrent policies. All cells, including failures and contrary estimates, are retained. A one-year availability lag is an assumption, not an observed filing calendar.

These contrasts measure market environments. Historical global issuer financials, fiscal years and quote-to-filer group links do not support a verified eligible-issuer treatment panel. The two banking phases overlap with each other and with European banking reforms. They are one collective assignment; country clustering cannot create independent policy variation. We therefore report descriptive magnitudes without causal liquidity p-values.

### 3.3 International implementation

The diffusion panel uses documented independent country-class-years. First-recorded-phase timing models end at the focal phase; status models examine recorded implementation. Common EU waves are excluded as focal independent choices, including both banking phases. Missing histories remain unknown, and documented scope or enforcement changes censor affected status observations.

Following Koning et al. (2018), competition uses lagged implementation among five export-pattern peers; learning uses prior relative GDP growth or liquidity among geographic neighbours; emulation uses language, legal-origin or IGO peers; external dependence uses IMF credit including SDR allocations and net aid relative to GDP. Signals precede the focal year. Models include family-year strata, country-clustered covariance and, in controlled designs, lagged GDP per capita, growth and trade/GDP. We retain the complete fixed matrix of logit and linear-probability specifications, support failures and diagnostics. Holm correction covers every stable focal coefficient rather than a selected favourable subset.

## 4 Empirical Results

### 4.1 Liquidity around public reporting

Table 2 reports the core comparisons. Around the 2014 initial banking phase, log spreads fall by 0.247 with equal-country weights and 0.109 with constant-GDP weights, equivalent to 21.9% and 10.3% relative spread differences. Fifteen treated markets support these cells. The GDP-weighted comparison has an effective treated-country count of 7.3, with Germany accounting for 22.1% of treated weight. The six weighting schemes preserve the negative sign but imply relative differences from about 1.0% to 44.6%; weighting materially changes the economic summary.

The 2015 full banking phase also has negative baseline log-spread comparisons: -0.113 under equal weights and -0.032 under GDP weights. These are overlapping calendar-window contrasts, not two independent policy experiments or an estimate of the incremental effect of publishing tax information. Figure 1 places the bank phases beside reviewed ESG, IFRS and confidential tax reporting using the same outcome definitions.

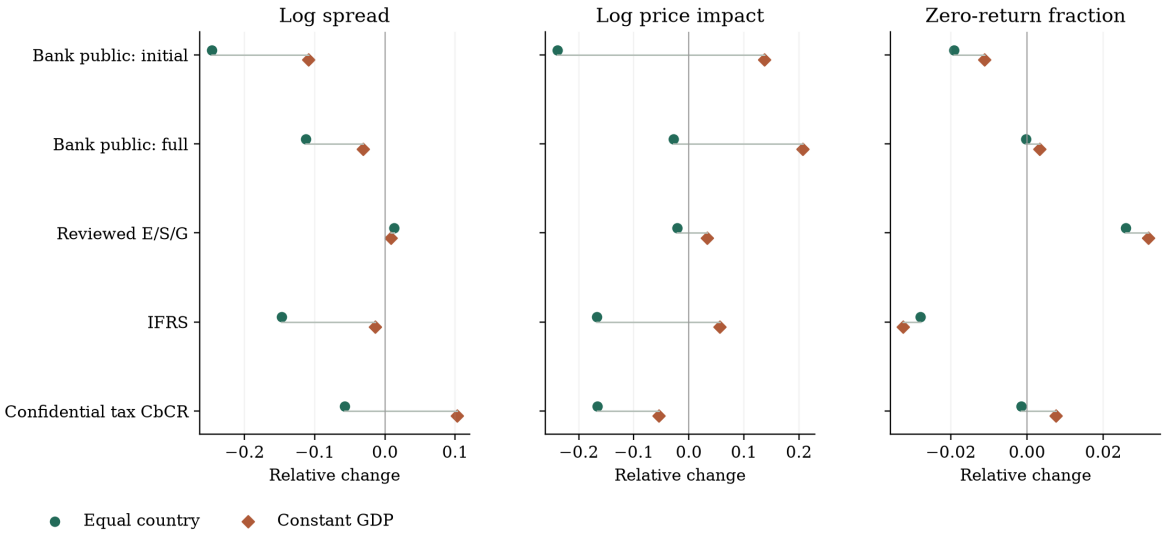
The evidence does not support an unrestricted liquidity claim. Initial-bank log price impact is -0.239 with equal-country weights but 0.137 with GDP weights. The corresponding full-phase values are -0.028 and 0.207. Reviewed ESG log-spread comparisons are 0.013 and 0.008, while zero-return fractions increase. Thus the strongest finding is a bank-phase spread association, not uniform improvement across liquidity proxies or reporting regimes.

**Table 2 Liquidity around reviewed reporting phases**

| Regime | Outcome | Equal country [N] | Constant GDP [N] |
|---|---|---|---|
| Bank public: initial | Log spread | -0.247 [15] | -0.109 [15] |
| Bank public: initial | Log impact | -0.239 [14] | 0.137 [14] |
| Bank public: initial | Zero-return fraction | -0.019 [15] | -0.011 [15] |
| Bank public: full | Log spread | -0.113 [15] | -0.032 [15] |
| Bank public: full | Log impact | -0.028 [14] | 0.207 [14] |
| Bank public: full | Zero-return fraction | -0.000 [15] | 0.003 [15] |
| Reviewed E/S/G | Log spread | 0.013 [23] | 0.008 [22] |
| Reviewed E/S/G | Log impact | -0.021 [23] | 0.033 [22] |
| Reviewed E/S/G | Zero-return fraction | 0.026 [23] | 0.032 [22] |
| IFRS | Log spread | -0.147 [20] | -0.014 [20] |
| IFRS | Log impact | -0.167 [21] | 0.056 [21] |
| IFRS | Zero-return fraction | -0.028 [22] | -0.033 [22] |
| Confidential tax CbCR | Log spread | -0.057 [42] | 0.103 [42] |
| Confidential tax CbCR | Log impact | -0.166 [42] | -0.054 [42] |
| Confidential tax CbCR | Zero-return fraction | -0.002 [43] | 0.008 [43] |

Changes from g-1 to g+1 use the same securities within country, at least 30 balanced quotes and at least 3 comparison countries. Brackets give treated-market counts. Negative values indicate lower spreads, impact or zero-return fractions. Bank rows compare EU versus non-EU home-geography market environments, not eligible bank issuers. GDP weights are measured in g-1; samples can differ with macro coverage. Both bank phases are one collective assignment and overlapping windows. These are descriptive contrasts; no causal p-values are reported.

**Figure 1 Reporting phases, weighting and liquidity**




Points are descriptive g+1 versus g-1 contrasts; connecting lines indicate different economic weights, not confidence intervals. Bank rows measure whole-market home-regime geography. Higher GDP-weighted price impact qualifies the negative bank spread comparisons.

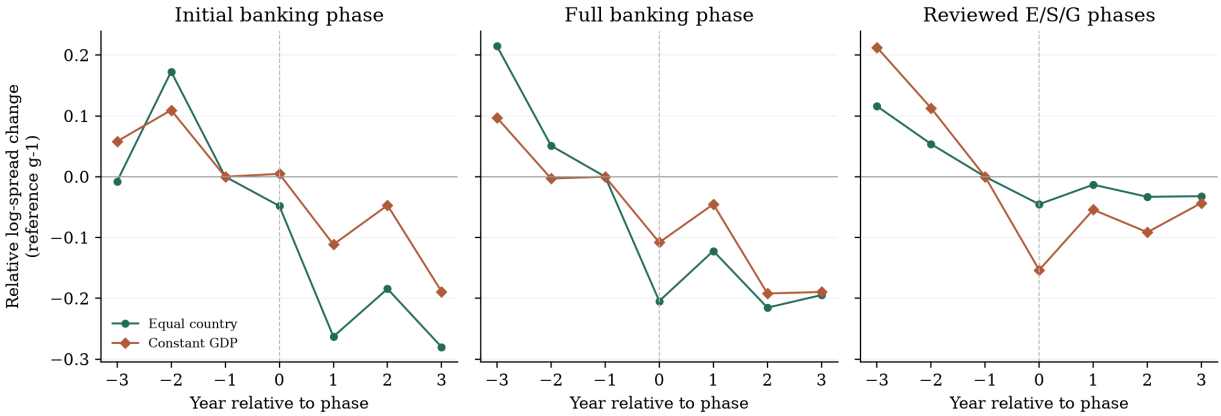
### 4.2 Timing and sensitivity

Figure 2 fixes the securities, countries and cohorts from g-3 through g+3. Initial-bank spread comparisons remain below their g-1 reference after the phase, but earlier relative movements and the broader European reform environment prevent a causal attribution. Balanced samples differ from the two-year baseline and are identified separately.

The initial banking spread comparison remains negative with a one-year availability lag, pretrend extrapolation, median quote changes, capitalisation weighting, common-outcome samples and alternative daily-data screens. The full-phase result is less stable: pretrend-adjusted spread comparisons are 0.085 with equal weights and 0.059 with GDP weights, reversing the baseline sign. Removing the collective EU assignment leaves no treated banking wave; concurrent-policy exclusions also remove the required bank comparison support. These diagnostics qualify the banking interpretation rather than establish an isolated effect.

The separate published ESG calendar produces lower spread and price-impact comparisons, but uses a different country sample, dates and covered populations. Replacing that calendar with reviewed legal phases changes more than dates alone. The difference cannot be attributed solely to a particular coding correction, nor can one calendar be chosen because it delivers a preferred sign. Complete benchmark and reviewed results are available together.

**Figure 2 Spread profiles on balanced seven-year samples**




Securities, countries and cohorts are held fixed from g-3 through g+3 within each panel/weight. The g-1 reference is zero by construction. Initial/full bank panels each have 13 treated markets; reviewed ESG has 12 equal-weight and 11 GDP-weighted treated markets. Bank windows overlap; earlier relative movements and concurrent reforms preclude a causal interpretation.

### 4.3 International diffusion

Table 3 presents the four focal timing associations. Export-peer exposure has an uncontrolled logit coefficient of 0.458 (raw p=0.088); adding economic controls changes its sign. Financial dependence is negatively associated with phase timing (-0.720, raw p=0.0003), opposite to a positive coercion prediction. Its broad Holm-adjusted p-value is 0.142. Neither foreign GDP-growth learning nor language-peer exposure provides precise evidence for the predicted positive channel.

Across 1,330 attempted designs, 433 stable focal coefficients remain after the stated support and stability screens; none passes the full-family Holm correction. This statement concerns statistical support, not proof that diffusion mechanisms are absent. Network coverage and known legal histories

limit the usable samples, and an IMF/aid association would not establish policy-specific pressure even if precisely estimated.

The banking wave is institutional evidence of coordinated implementation through EU law. It is not evidence that 28 countries independently learned from prior liquidity gains. The timing models therefore cannot turn the observed bank spread pattern into a learning mechanism. Linking the two parts requires prior, credibly identified foreign benefits and independent subsequent domestic choices.

**Table 3 Independent phase timing and diffusion measures**

| Measure | Controls | Logit coefficient (SE) | Raw p | Broad Holm p | Rows / countries |
|---|---|---|---|---|---|
| Export peers | No | 0.458 (0.264) | 0.0883 | 1.000 | 233/58 |
| Export peers | Yes | -0.055 (0.320) | 0.8654 | 1.000 | 218/55 |
| Relative GDP-growth learning | No | -0.409 (0.695) | 0.5601 | 1.000 | 74/33 |
| Relative GDP-growth learning | Yes | -0.754 (0.850) | 0.3821 | 1.000 | 69/31 |
| Language peers | No | 0.105 (0.275) | 0.7032 | 1.000 | 411/66 |
| Language peers | Yes | 0.009 (0.281) | 0.9737 | 1.000 | 348/53 |
| Credit + aid / GDP | No | -0.720 (0.186) | 0.0003 | 0.142 | 269/48 |
| Credit + aid / GDP | Yes | -0.798 (0.345) | 0.0257 | 1.000 | 255/45 |

Family-year strata and country-clustered covariance; t reference with countries-1 degrees of freedom. Continuous focal variables are standardised within each model; relative-growth learning is binary. Controls are lagged GDP per capita, growth and trade/GDP. Broad Holm correction covers all 433 stable focal coefficients in the complete matrix, including status/LPM alternatives. No broad-adjustment discovery remains. Common EU banking waves do not enter as independent national events. Sample support and unavailable/unstable fits are retained online.

## 5 Conclusion

Public reporting rules are differentiated information interventions. The clearest descriptive result is lower country-market spreads around the initial European banking transparency phase, with a consistent direction across economic weights and several diagnostics. The result is narrower than a general capital-market benefit: price impact, the full-reporting timing checks and the reviewed ESG phases do not support the same unrestricted conclusion.

Mandatory topic responses, covered populations and public filing dates must be aligned before moving from national-market associations to issuer effects. Coordinated implementation must likewise be separated from independent diffusion. The current evidence motivates a focused bank-transparency inquiry, while a causal bank-issuer effect and a liquidity-driven diffusion mechanism remain unestablished.

## References

Amihud, Y. 2002. Illiquidity and stock returns: Cross-section and time-series effects. Journal of Financial Markets 5:31-56.

Diamond, D. W., and R. E. Verrecchia. 1991. Disclosure, Liquidity, and the Cost of Capital. Journal of Finance 46:1325-1359.

Holm, S. 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6:65-70.

Ince, O. S., and R. B. Porter. 2006. Individual Equity Return Data from Thomson Datastream: Handle with Care! Journal of Financial Research 29:463-479.

Koning, M., G. Mertens, and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies 49:249-271.

Krueger, P., Z. Sautner, D. Y. Tang, and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. January working paper, ECGI Finance Working Paper 754/2021.

European Union. 2013. Directive 2013/36/EU, Article 89. European Union. 2014. Directive 2014/95/EU, Article 4.

HM Treasury. Capital Requirements (Country-by-Country Reporting) Regulations 2013: guidance. SI 2013/3118, regulations 2-5.

France. Law 2001-420, Article 116. Code monetaire et financier, L511-45, version 28 July 2013. Ireland. SI 360/2017, regulation 2. Finland. Act 1376/2016, transition. Greece. Law 4403/2016 and circular 62784/7 June 2017.

European Banking Authority. 2014 EU-wide stress-test bank list; 2015 EU-wide transparency bank list. HSBC Holdings plc. Country-by-Country Reporting 2013 and 2014; historical reporting index. Deutsche Bank. Financial Report 2014, note 45. BNP Paribas Fortis. First-half 2014 and Annual Report 2014, section 3.c.

Barbieri, K., and O. M. G. Keshk. 2016. Correlates of War Trade Data, Version 4.0. Mayer, T., and S. Zignago. 2011. The GeoDist database. CEPII 2011-25. Pevehouse, J. C., et al. 2019. IGOs dataset. Journal of Peace Research 57:492-503.

## Data availability

Aggregate results, legal coverage rules, individual policy-review JSONs and source references: https://hunternbh.github.io/policy-diffusion/. Licensed market observations and archived third-party originals are retained locally. Remaining evidence gaps are explicit in the companion materials.